

Nimbus - V2

Preliminary Design Review -PDR

**Presented To:
University of California, San Diego
Department of Mechanical and Aerospace Engineering**

**MAE 155B
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1. Executive Summary

The Nimbus V2 is a radio-controlled aircraft designed to efficiently deliver payload while maximizing mission profit under aerodynamic, propulsion, and structural constraints. A flying-wing configuration was selected for its superior aerodynamic efficiency, achieving an effective lift-to-drag ratio of 19.61 and improved payload volume utilization.

The aircraft has a wingspan of 1.33 m, wing area of 0.208 m², and gross weight of 20.06 N, with a design wing loading of 96.29 N/m² and required thrust of 12.15 N ($T/W = 0.605$). The propulsion system, consisting of a SunnySky X2216 motor, 3S LiPo battery, and APC 10×4.7 propeller, provides 17.77 N of static thrust while remaining within current limits. Mission analysis predicts an energy requirement of 8.87 Wh (including reserve) over a 1.46 km mission, yielding a profit rate of 1.70 \$/hr.

The structure uses a composite and hybrid wing design with a carbon fiber spar, achieving a factor of safety of 5.88 while meeting acceptable deflection limits. Overall, the design meets key performance and structural requirements and provides a viable baseline for further refinement.

Table 1: Aircraft Specifications Summary

Parameter	Value
Empty Weight W_e	9.78 N
Wing Area S	0.2083 m ²
Wingspan b	1.331 m
Aspect Ratio AR	8.50
Wing Loading W/S	96.29 N/m ²
Lift-to-Drag Ratio L/D	19.61
Design Lift Coefficient C_L	0.4046
Parasite Drag Coefficient C_{D0}	0.0300
Oswald Efficiency Factor e	0.80
Required Thrust	12.15 N
Thrust-to-Weight Ratio T/W	0.605
Static Thrust	17.77 N

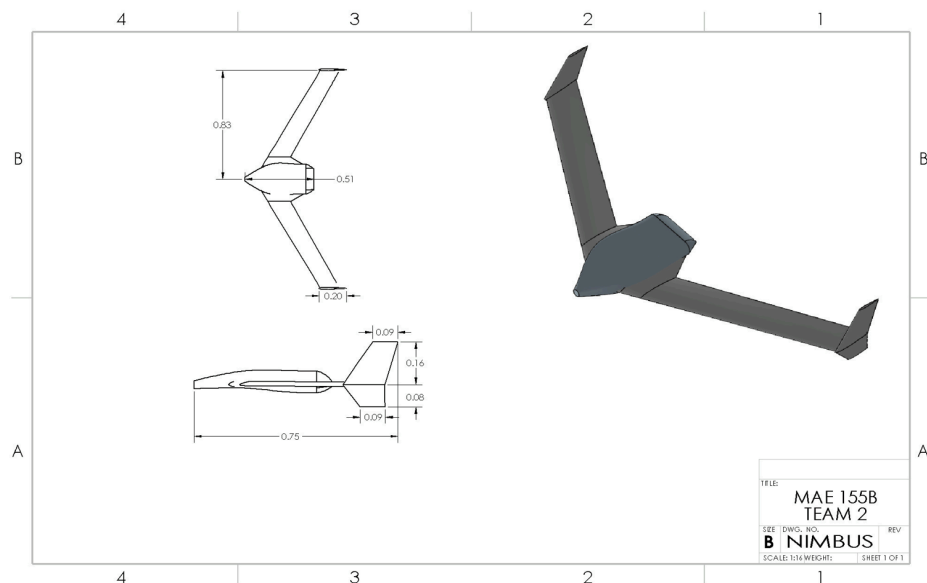


Figure 1: Technical Drawing

2. Introduction

The objective of this project is to design and develop a radio-controlled aircraft capable of efficiently delivering a payload while maximizing performance based on a defined scoring function. This design process considers structural integrity, aerodynamic efficiency, propulsion, and mission requirements, using analytical methods and MATLAB-based modeling to guide decision-making. Multiple configurations were evaluated, and a flying-wing layout was selected for its superior lift-to-drag ratio. Moreover, this configuration efficiently uses the internal volume. Although this configuration introduces challenges to stability and structural design, these are addressed through careful sizing, placement of the center of gravity, and structural reinforcement. This report presents the preliminary design, including configuration selection, initial sizing, and key design considerations for the proposed aircraft.

3. Scoring Function Analysis

The optimization results indicate that the objective function is strongly dominated by the volume scaling parameter, with the best performance achieved at the upper bounds of both design variables. The optimal configuration corresponds to a payload weight of 1000 g and a volume scalar $VPS = 1.10$, yielding a performance metric of $J = 0.00740$ \$/s (26.63 \$/hr). Under these conditions, the aircraft operates with an effective lift-to-drag ratio of 14.81 and an empty-weight fraction of 0.556, resulting in a derived gross weight of approximately 2.81 kg. The local sensitivity analysis further reinforces this trend, showing that the objective function is nearly an order of magnitude more sensitive to changes in VPS (0.992) than to payload weight (0.121). This indicates that aerodynamic and volumetric scaling effects dominate system performance, while increases in payload weight provide comparatively limited gains. Overall, the results suggest that maximizing volume efficiency is the primary driver for improving mission performance within the explored design space.

4. Initial Sizing

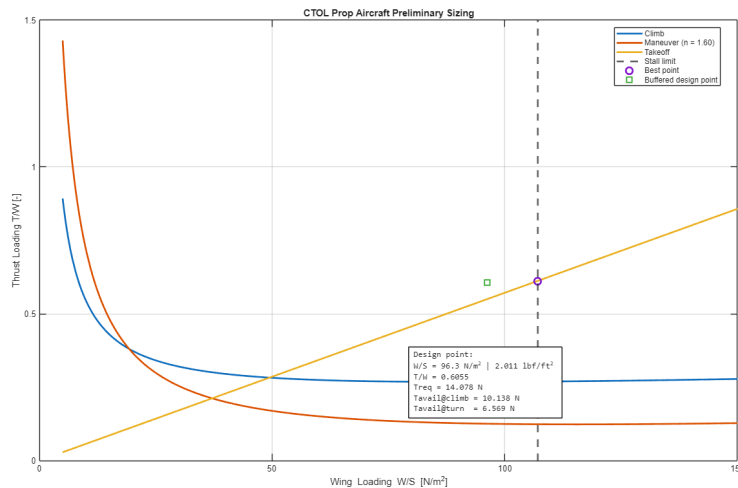


Figure 2: Preliminary Sizing

The initial sizing of the aircraft was performed using constraint analysis to determine an appropriate design point that satisfies performance requirements for takeoff, climb, and cruise. A

trade-off between thrust-to-weight ratio (T/W) and wind loading (W/S) was evaluated, where key constraints such as stall speed, climb performance, and propulsive limits were plotted to identify feasible design regions. The selected design point is in the overlap of these constraints and provides sufficient lift at low speeds and sufficient thrust for climb and maneuvering. This approach allows the Nimbus to reach a balanced configuration that meets the aerodynamic performance requirements without exceeding the structural or propulsion limits. The resultant wing loading and thrust-to-weight ratio are the basis for aerodynamic, structural, and propulsion design considerations.

5. Concept

Table 2: Configuration Selection

Criterion	Conventional	Flying Wing
Aerodynamic efficiency	-	++
Structural efficiency	-	+
Payload volume efficiency	-	+
Energy efficiency	-	++
Static stability	++	-
Control complexity	++	-
Manufacturing simplicity	+	-
Total	1	3

Conventional and flying wing configurations are initially considered. The flying wing configuration offers significant advantages in aerodynamic and structural efficiency, due to the superior lift-to-drag ratio, improving energy efficiency and overall performance. Additionally, the flying wing provides more efficient use of internal volume, enabling a preferred payload integration. While the conventional configuration offers superior static stability and simpler controls, these benefits primarily reduce design risk rather than improving mission performance.

6. Aerodynamics

The aerodynamic analysis models aircraft performance using a finite-wing lift-curve relation and quadratic drag polar developed from course methods. Two-dimensional airfoil data for the E222 root airfoil and E230 tip airfoil were obtained using XFOIL. These two-dimensional results were used to produce the three-dimensional wing values using the finite-wing lift-slope relation and induced-drag formulation. Figure 3, the three-dimensional coefficient of lift (CL) changes with a change in angle of attack (α), indicating the required cruise lift coefficient of $CL = 0.405$ occurring at an angle of attack of 2 degrees. Figure 4 shows the drag polar generated from the sum of parasite drag and induced drag, where parasite drag was estimated using the drag build-up method and induced drag was calculated using an Oswald efficient factor of $e = 0.80$. The analysis predicts a minimum drag coefficient of $CD_0 = 0.030$ and an effective lift-to-drag ratio near 19.6. Overall, the E222 root airfoil provides strong lift capability, while the E230 tip airfoil improves spanwise loading and stall behavior, making the flying wing configuration well-suited for the package-delivery mission.

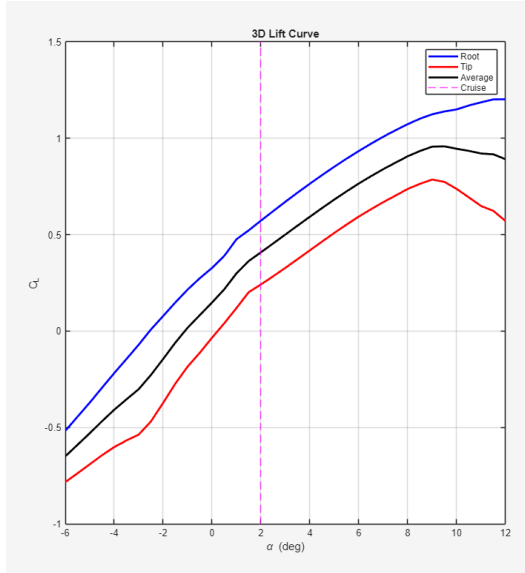


Figure 3: Lift Coefficient (C_L) versus Angle of Attack (α) for root airfoil (blue), tip airfoil (red), and average (black)

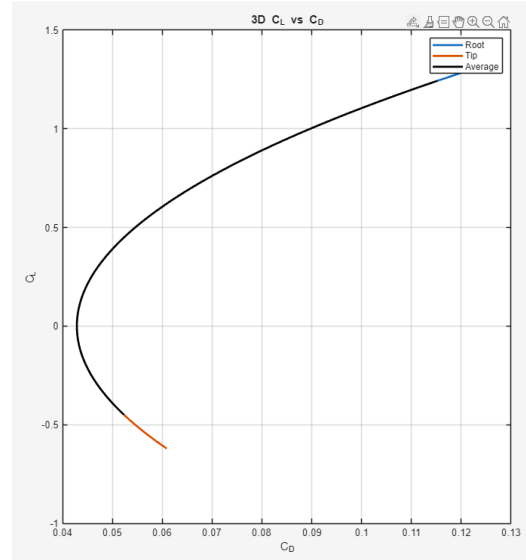


Figure 4: Drag Polar: Lift Coefficient (C_L) versus Drag Coefficient (C_D) for root airfoil (blue), tip airfoil (red), and average (black)

7. Propulsion

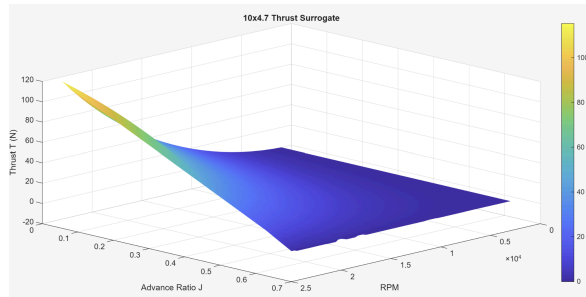


Figure 5: Thrust surrogate model for APC 10x4.7 Propeller

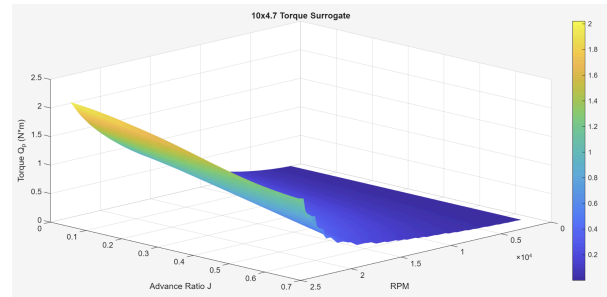


Figure 6: Torque surrogate model for APC 10x4.7 Propeller

Nimbus's propulsion system consists of a brushless electric motor, an electronic speed controller (ESC), a 3S 11.1 V LiPo battery, and an APC 10x4.7 propeller, selected to provide the thrust required. The system uses the SunnySky X2216 V3 1100 KV motor, paired with a 40A ESC. Motor behavior is modeled using steady-state relationships between voltage, current, and angular speed, while propeller behavior is represented using APC performance data parametrized by rotational speed and advance ratio.^[1]

Several candidate propellers were compared using thrust capability, efficiency, and current draw as selection criteria. The APC 10x4.7 propeller was selected due to a favorable balance between thrust production and torque demand. The propeller provides a strong low-speed thrust while avoiding excessive torque that would drive current beyond the desired operating range.

8.1. Structures and Materials

Table 3: Properties of PA6-GF

Property	Value	Units
Density	1300	kg/m ³
Tensile Strength	200	MPa
Young's Modulus	12	GPa
Flexural Strength	300	MPa
Heat Deflection Temperature	182	°C

Nimbus's structure combines Nylon-6 reinforced with glass fiber (PA6-GF), balsa wood, and styrofoam to balance structural integrity, thermal resistance, and manufacturability. The fuselage is constructed from PA6-GF, a composite material known for its stiffness, strength, and thermal stability. This material is suitable for electronic housing due to its heat deflection temperature, ensuring stability during flight. The wing structure consists of a hybrid approach to minimize weight while maintaining stiffness and load capacity. The primary aerodynamic surface is made from styrofoam, providing low density and simplifying shaping. Balsa wood ribs are integrated from within the wing to preserve airfoil geometry and distribute loads efficiently. A carbon fiber spar serves as the primary structural member, carrying bending loads caused by lift. Carbon fiber is selected for its high stiffness-to-weight ratio, minimizing deflection under aerodynamic loads. This combination results in a lightweight, structurally efficient wing design.

8.2 V-n Diagram

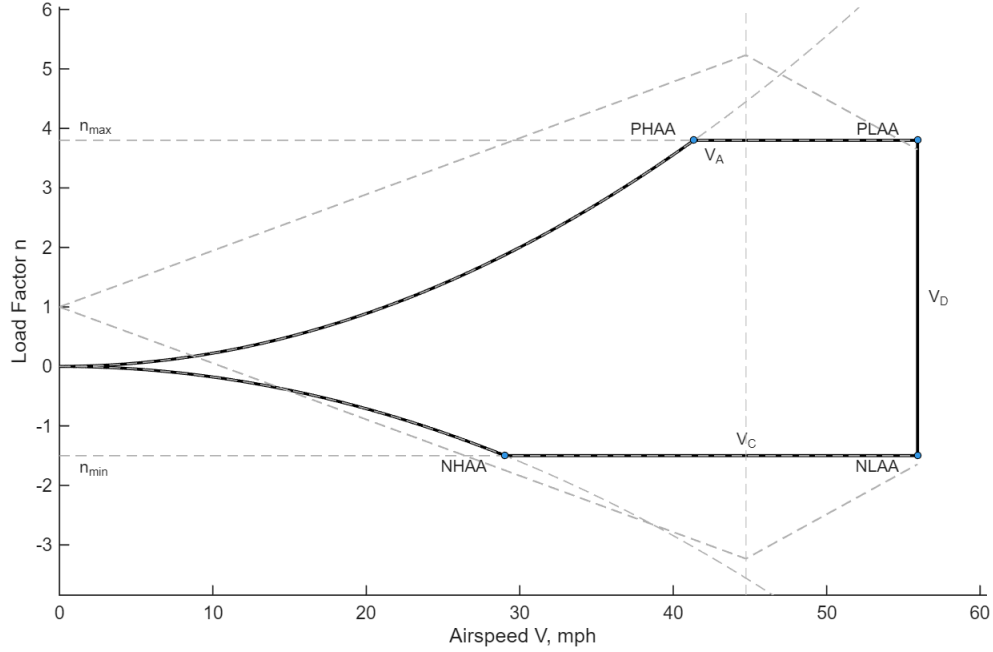


Figure 9: V-n Diagram (Maneuver and Gust Load Envelope)

The V-n diagram depicts the preliminary combined maneuver and gust load for Nimbus. The maneuver envelope was constructed using the current gross weight W_g , the sized wing reference area S_{ref} , density $\rho = 1.19 \text{ kg/m}^3$, and $C_{L,max} = 1.8$. S_{ref} is obtained directly from the CTOL sizing output through $S_{ref} = W_g / (W/S)_{design}$, while the cruise condition is based on

the mission pattern speed $V_c = 20 \text{ m/s}$ ($\approx 44.7 \text{ mph}$). The structural design limits were taken as $n_{max} = 3.8$ and $n_{min} = -1.5$, and the dive speed was estimated as $V_D = 1.25 * V_c = 25 \text{ m/s}$ ($\approx 55.9 \text{ mph}$), as a design assumption. The positive and negative stall boundaries were computed from the lift equation, with the negative boundary based on the additional assumption $C_{L,max,-} = -0.8C_{L,max}$. The dashed gust lines were overlaid to estimate the additional load-factor excursions produced by vertical gusts and to assess whether gust loading may become more restrictive than maneuver loading at higher speeds. Since several of these quantities, including $C_{L,max}$, V_D , the negative lift limit and the gust inputs, are preliminary assumptions, the figure should be interpreted as a first-pass structural envelope rather than a final certification-level result; however, it is still valuable for identifying the critical speed range and the likely governing load cases early in the design process.

9. Static Stability

The Nimbus V2 remains longitudinally statically stable, maintaining a static margin of approximately (18 to 19)% MAC depending on the payload configuration. The aircraft exhibits a static margin of 18.59% MAC with payload and 19.81% MAC without payload, indicating that payload deployment produces only a 1.22% change in static margin. This small shift demonstrates that the vehicle's mass distribution effectively limits center-of-gravity migration during payload release by placing the heaviest components near the aft fuselage at the calculated quarter MAC. As a result, the aircraft maintains positive static longitudinal stability throughout the mission profile.

9. References

- [1] APC Propellers, “PER3_10x47SF Propeller Data File,” *APC Propeller Performance Database*. [Online]. Available: http://www.apcprop.com/files/PER3_10x47SF.dat. [Accessed: Apr. 21, 2026].

10. Appendix

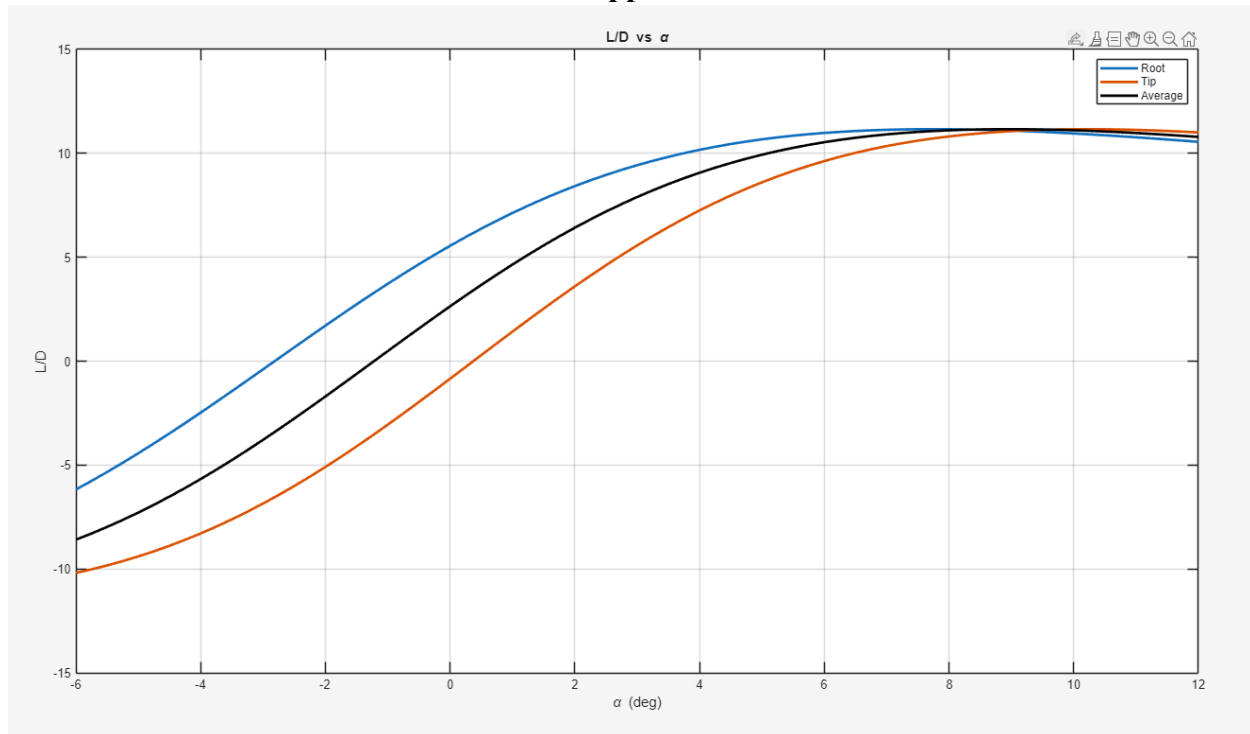


Figure A1: Lift to drag ratio (L/D) versus angle of attack for root airfoil (blue), tip airfoil (red), and average (black)

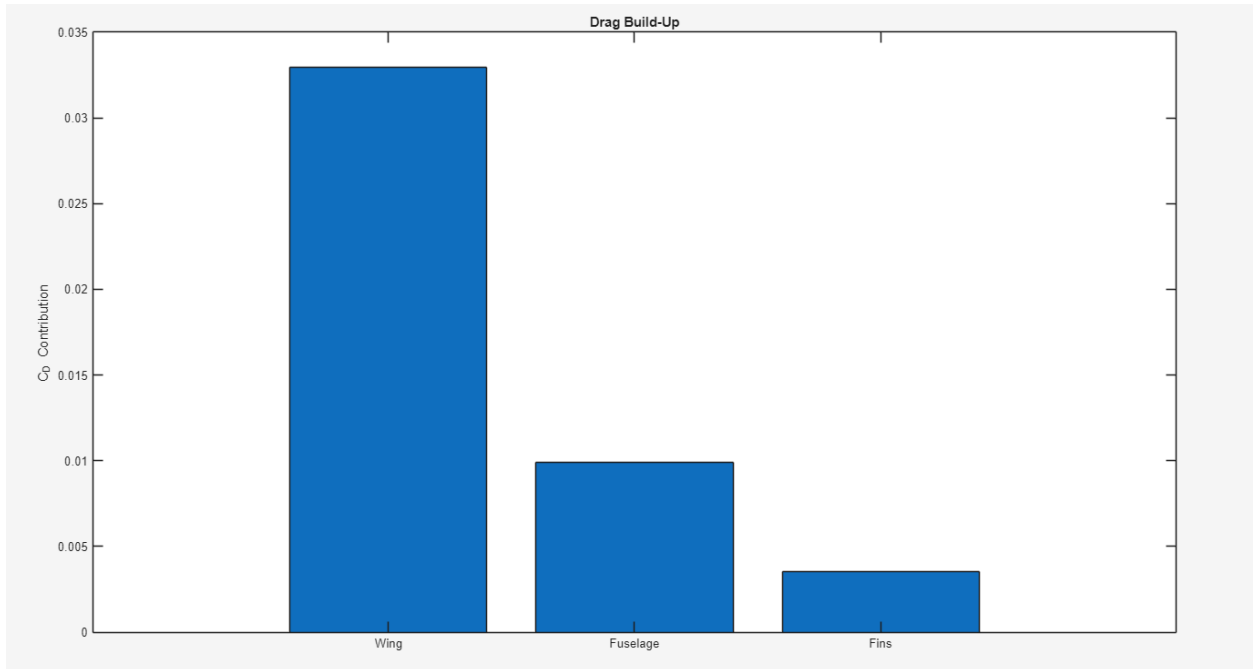


Figure A2: Drag build-up bar plot representing drag contributions from wing, fuselage, and fins.

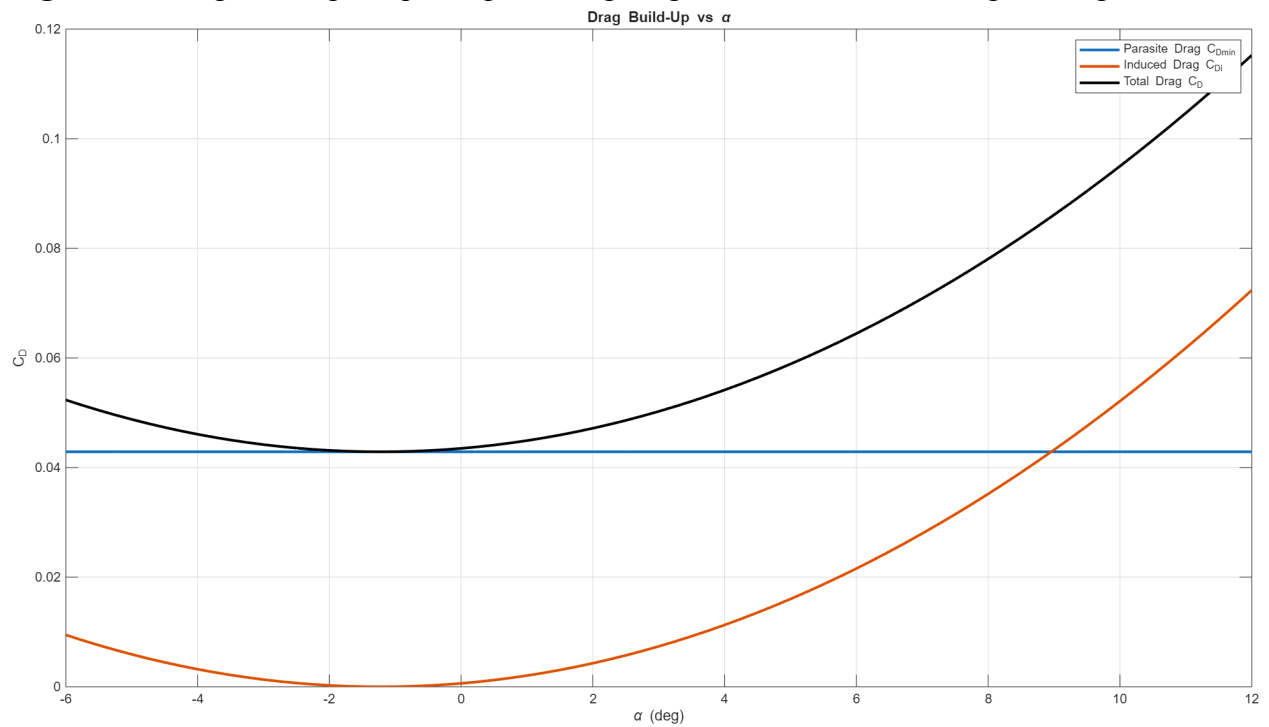


Figure A3: Drag build-up versus alpha showing parasite drag (blue), induced drag (red), and total drag (red)

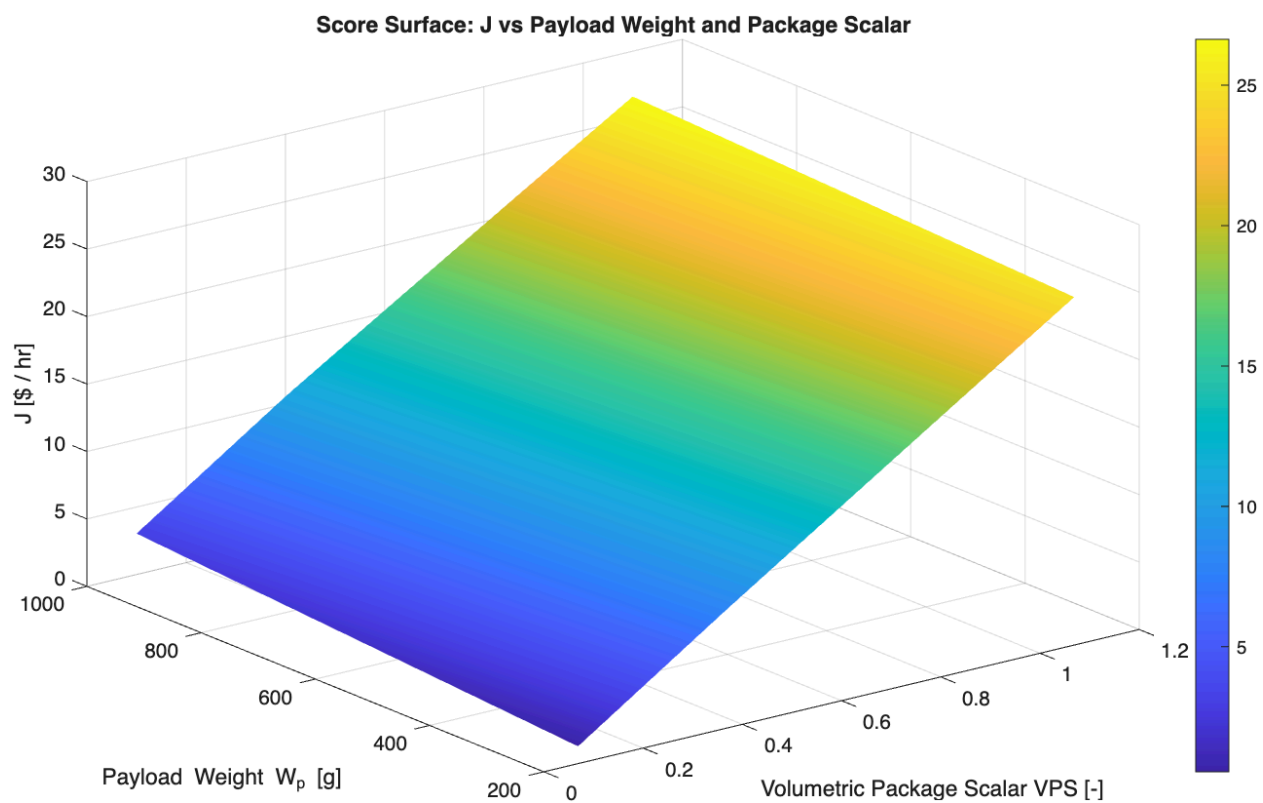


Figure A4: Score Function Analysis Optimization